

INFERENCEAL STATISTICS

PROJECT

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Problem 1:

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Position	Striker	Forward	Attacking Midfielder	Winger	Total
0	Players Injured	45	56	24	20	145
1	Players Not Injured	32	38	11	9	90
2	Total	77	94	35	29	235

Figure 1 physiotherapist with a male football team players data

1.1 What is the probability that a randomly chosen player would suffer an injury?

The probability of an event is calculated as:

$$\Rightarrow P(A) = \frac{\text{No of Favourable outcomes}}{\text{Total Number of outcomes}}$$

- Number of injured players = 145
- Total Number of players = 235

$$\Rightarrow P(\text{injury}) = \frac{\text{No of injured players}}{\text{Total Number of players}}$$

$$\Rightarrow P(\text{injury}) = \frac{145}{235}$$

$$\Rightarrow P(\text{injury}) \approx 0.617$$

Conclusion:

So, the probability that a randomly chosen player would suffer an injury is approximately 0.617 or 61.7 %.

1.2 What is the probability that a player is a forward or a winger?

For finding the probability for that a player is forward or winger, we need to find the probability of forward player and then we will find the probability of winger player and we use addition rule

- Total number of players = 235
- Number of Winger = 29
- Number of forward = 94

$$\Rightarrow P(\text{Forward}) = \frac{\text{Number of forward}}{\text{Total Number of Players}}$$

$$\Rightarrow P(\text{Forward}) = \frac{94}{235}$$

$$\Rightarrow P(\text{Winger}) = \frac{\text{No of Winger}}{\text{Total Number of Players}}$$

$$\Rightarrow P(\text{Winger}) = \frac{29}{235}$$

$$\Rightarrow P(\text{Forward} \cup \text{Winger}) = P(\text{Forward}) + P(\text{Winger})$$

$$\Rightarrow P(\text{Forward} \cup \text{Winger}) = \frac{94}{235} + \frac{29}{235}$$

$$\Rightarrow P(\text{Forward} \cup \text{Winger}) = \frac{123}{235}$$

$$\Rightarrow P(\text{Forward} \cup \text{Winger}) \approx 0.523$$

Conclusion:

So, the probability that a player is a forward or a winger are approximately 0.523 or 52.3%

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

We will calculate the probability that a randomly chosen player plays in a striker position and has a foot injury is the no of players who plays in striker position divided by Total number of players

- No of striker who is having foot injury = 45
- Total Number of players = 235

$$\Rightarrow P(\text{Probability of striker with foot injury}) = \frac{\text{No of striker who is having foot injury}}{\text{Total Number of players}}$$

$$\Rightarrow P(\text{Probability of striker with foot injury}) = \frac{45}{235}$$

$$\Rightarrow P(\text{Probability of striker with foot injury}) \approx 0.191$$

Conclusion:

The probability that a randomly chosen player plays in a striker position and has a foot injury are approximately 0.191 or 19.1 %.

1.4 What is the probability that a randomly chosen injured player is a striker?

For finding the probability that a randomly chosen injured player is a striker is total number of players who are injured in striker position divided by the total number of injured player.

- Total number of injured strikers = 145
 - Number of strikers with foot injury = 45
- $$\Rightarrow P(\text{being striker given Injury}) = \frac{\text{Number of strikers with foot injury}}{\text{Total number of injured strikers}}$$

$$\Rightarrow P(\text{being striker given Injury}) = \frac{45}{145}$$

$$\Rightarrow P(\text{being striker given Injury}) \approx 0.310$$

Conclusion:

So, the probability that a randomly chosen injured player is a striker is approximately 0.310 or 31.0%

Problem 2:

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimetre and a standard deviation of 1.5 kg per sq. centimetre. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain; Answer the questions below based on the given information; (Provide an appropriate visual representation of your answers, without which marks will be deducted)

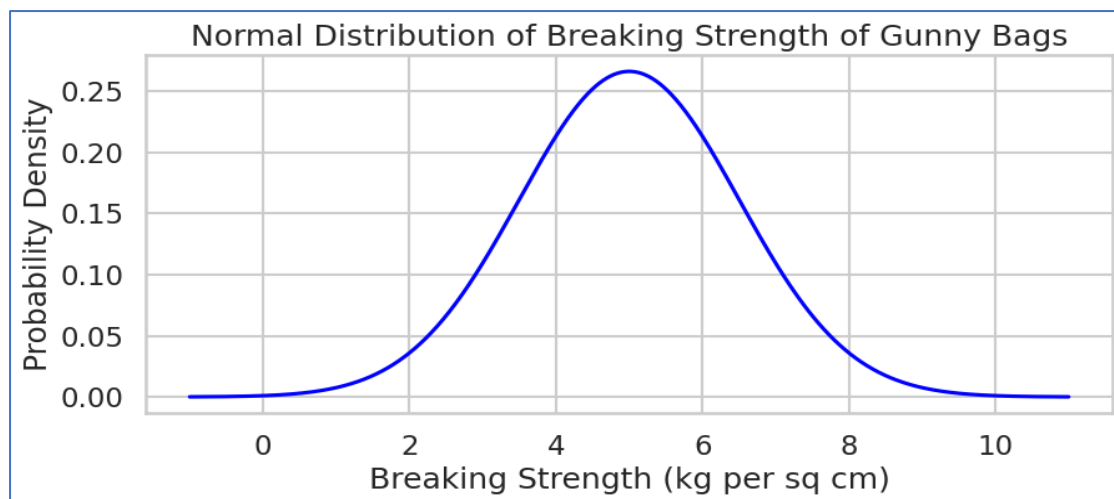


Figure 2 Normal distribution of Breaking strength of gunny bags

Observations:

- The breaking strength of gunny bags used for packaging cement is normally distributed as it is showing very well bel curved.

We need to use the properties of normal distribution and the Z- score:

$$\Rightarrow Z - Score = \frac{X - \mu}{\sigma}$$

- X = The given X value for calculation of statistic
- μ = Mean of population
- σ = Standard Deviation of population

In above problem, the standard deviation of length of gunny bags and mean of length of gunny bags are given:

Given That:

$\mu = 5$ Kg per sq. cm
 $\sigma = 1.5$ kg per sq.cm

2.1 What proportion of the gunny bags have a breaking strength less than 3.17 kg per sq. cm?

First, we will calculate the Z-score for 3.17 kg per sq. cm:

$$\Rightarrow Z - Score = \frac{X - \mu}{\sigma}$$

$X = 3.17$ kg per sq. cm
 $\mu = 5$ Kg per sq. cm
 $\sigma = 1.5$ kg per sq.cm

$$\Rightarrow Z - Score = \frac{3.17 - 5}{1.5}$$
$$\Rightarrow Z - Score = \frac{-1.83}{1.5}$$
$$\Rightarrow Z - Score \approx -1.22$$

Now by using standard normal distribution, we find the proportion of gunny bags corresponding to the Z-score of -1.22.

$$\Rightarrow P(X < 3.17) \approx 0.1112$$

From the Plot:

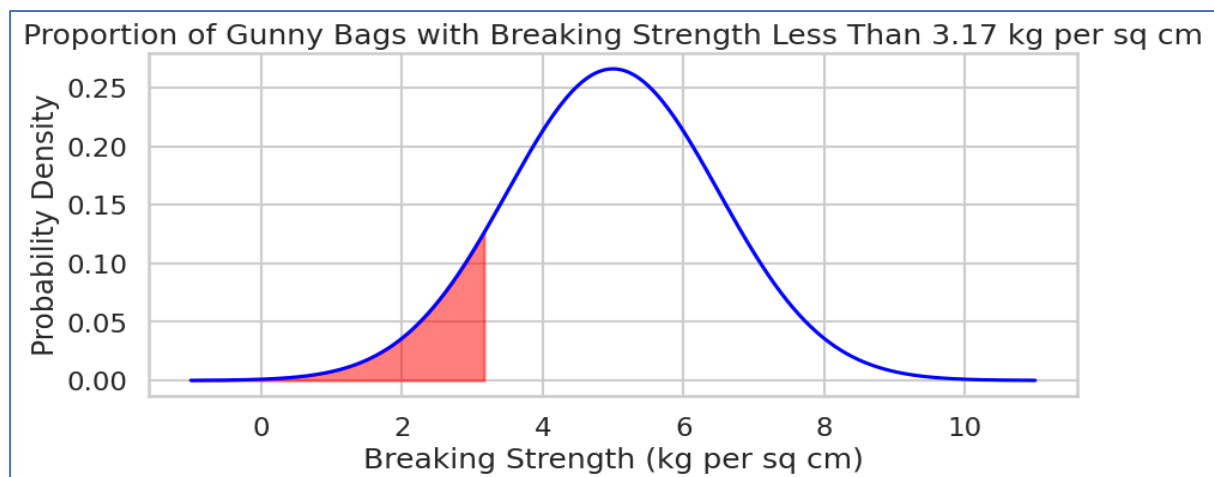


Figure 3 Proportion of gunny bags with breaking strength less than 3.17 kg per sq cm

Conclusion:

- With the help of Cumulative distribution Function (CDF), we had find the breaking strength which is less than 3.17 kg per sq cm.
- After analysing from graphically, we have found out the probability or area on the left side of red line are less than 3.17 kg per sq. cm. The cumulative Distribution on left side of line is 0.1112. Hence, the probability is 11.12%.

- Proportion of gunny bags with breaking strength less than 3.17 kg per sq. cm are approximately 0.1112 or 11.12%.

2.2 What proportion of the gunny bags have a breaking strength at least 3.6 kg per sq cm.?

First, we will calculate the Z-score for 3.16 kg per sq. cm:

$$\Rightarrow Z - Score = \frac{X - \mu}{\sigma}$$

$X = 3.6$ kg per sq. cm

$\mu = 5$ Kg per sq. cm

$\sigma = 1.5$ kg per sq.cm

$$\Rightarrow Z - Score = \frac{3.6 - 5}{1.5}$$

$$\Rightarrow Z - Score = \frac{-1.4}{1.5}$$

$$\Rightarrow Z - Score \approx -0.93$$

Using the Z-table, the proportion corresponding to a Z-score of -0.93 is about 0.1762. The proportion for at least 3.6 kg per sq. cm is:

$$\Rightarrow P(X \geq 3.6) = 1 - P(X < 3.6) = 1 - 0.1762 \approx 0.8247$$

Proportion of gunny bags with breaking strength at least 3.6 kg per sq cm: 0.8247 or 82.47%

From the Plot:

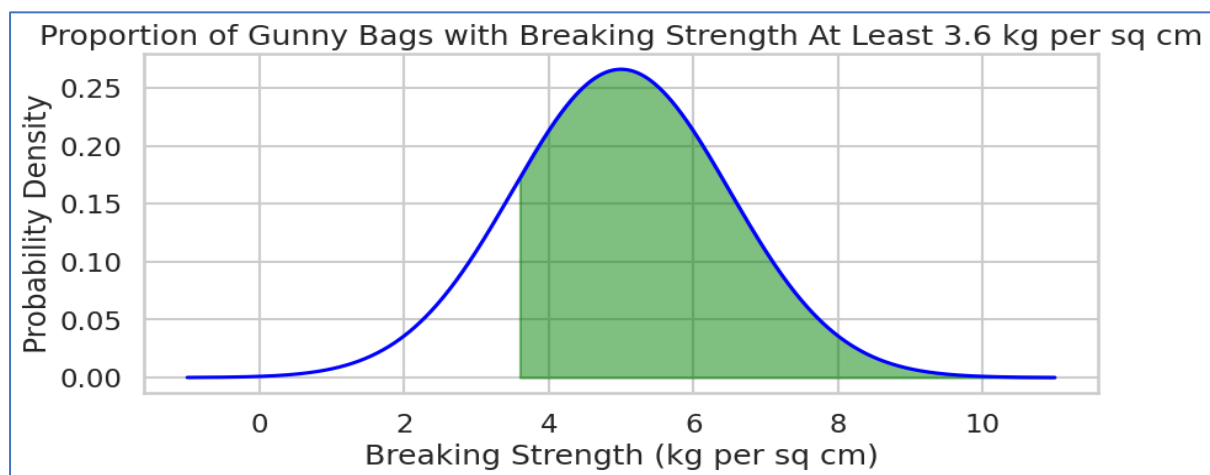


Figure 4 Proportion of gunny bags with breaking strength at least 3.6 kg per sq cm

Conclusion:

- We are using Cumulative Distribution Function (CDF) for finding the breaking strength at least 3.6 kg per sq. cm.
- After analysing the graph for the above question, we got the probability or area on the right side of the green line that the proportion is at least 3.6 kg per sq cm.

- The Cumulative Distribution area under the Red Region is 0.8247. So, the cumulative probability is 82.47%

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq. cm.?

Calculate the Z-score for 5 and 5.5 kg

$$\Rightarrow Z - Score = \frac{X - \mu}{\sigma}$$

$$X_1 = 5 \text{ kg per sq. cm}$$

$$X_2 = 5.5 \text{ kg per sq. cm}$$

$$\mu = 5 \text{ Kg per sq. cm}$$

$$\sigma = 1.5 \text{ kg per sq.cm}$$

$$\Rightarrow Z_1 - Score = \frac{5 - 5}{1.5}$$

$$\Rightarrow Z_1 - Score = \frac{0}{1.5}$$

$$\Rightarrow Z_1 - Score = 0$$

$$\Rightarrow Z_2 - Score = \frac{5.5 - 5}{1.5}$$

$$\Rightarrow Z_2 - Score = \frac{0.5}{1.5}$$

$$\Rightarrow Z_2 - Score \approx 0.33$$

Using the Z-table, the proportion corresponding to a Z-score of 5 and 5.5 are 0 and 0.33 respectively. The proportion between 5 and 5.5 kg per sq. cm are:

$$\Rightarrow P(5 \leq X \leq 5.5) = P(X \leq 5.5) - P(X \leq 5) = 0.6293 - 0.5 \approx 0.1293$$

Proportion of gunny bags with breaking strength between 5 and 5.5 kg per sq. cm: 0.1306 or approximately 13 %

From the Plot:

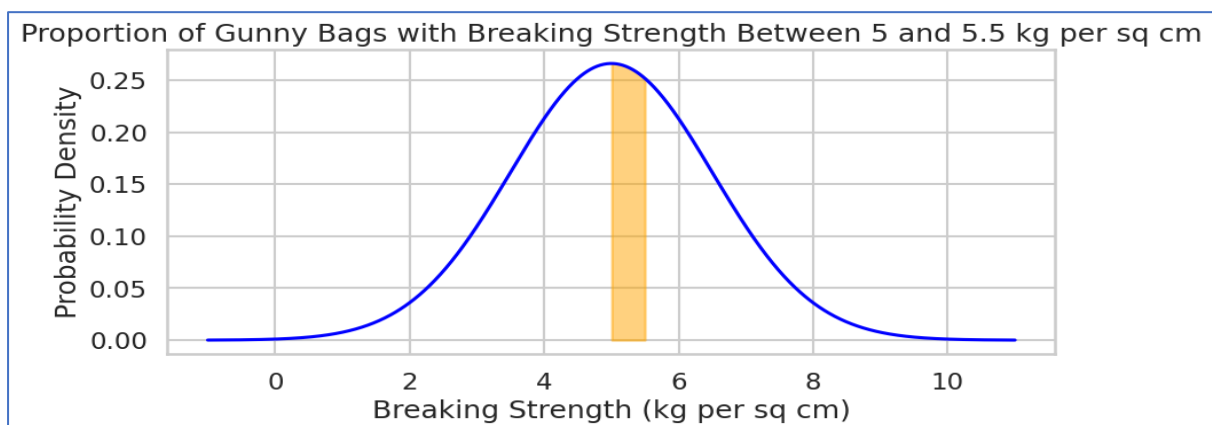


Figure 5 Proportion of gunny bags with breaking strength between 5 and 5.5 kg per sq cm

Conclusion:

- We are using Cumulative Distribution Function (CDF) for finding the breaking strength between 5 and 5.5 kg per sq. cm
- We need to find out the probability or area under the coloured lines for the given question that proportion between 5 and 5.5 kg per sq. cm.
- We can see that the cumulative probability is 13.1% which is low because the interval between 5 and 5.5 kg per sq. cm is very less.

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq. cm.?

Calculate the Z-score for 3 and 7.5 kg

$$\Rightarrow Z - \text{Score} = \frac{X - \mu}{\sigma}$$

$$X_1 = 3 \text{ kg per sq. cm}$$

$$X_2 = 7.5 \text{ kg per sq. cm}$$

$$\mu = 5 \text{ Kg per sq. cm}$$

$$\sigma = 1.5 \text{ kg per sq.cm}$$

$$\Rightarrow Z_1 - \text{Score} = \frac{3 - 5}{1.5}$$

$$\Rightarrow Z_1 - \text{Score} = \frac{-2}{1.5}$$

$$\Rightarrow Z_1 - \text{Score} \approx -1.33$$

$$\Rightarrow Z_2 - \text{Score} = \frac{7.5 - 5}{1.5}$$

$$\Rightarrow Z_2 - \text{Score} = \frac{2.5}{1.5}$$

$$\Rightarrow Z_2 - \text{Score} \approx 1.67$$

Using the Z-table, the proportion corresponding to a Z-score of -1.33 is about 0.0918 and for Z-score of 1.67 is about 0.9525. The proportion between 3 and 7.5 kg per sq cm are:

$$\Rightarrow P(3 \leq X \leq 7.5) = P(X \leq 7.5) - P(X \leq 3) = 0.9525 - 0.0918 \approx 0.8607$$

The proportion NOT between 3 and 7.5 kg per sq. cm is:

$$\Rightarrow 1 - P(3 \leq X \leq 7.5) = 1 - 0.8607 = 0.1393$$

Proportion of gunny bags with breaking strength NOT between 3 and 7.5 kg per sq cm: 0.1390 or 13 %.

From the Plot:

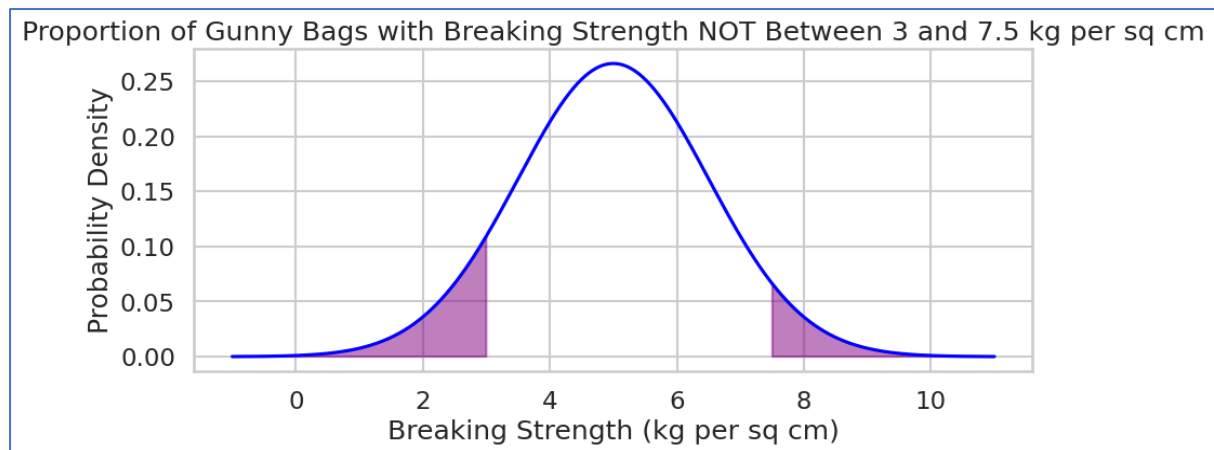


Figure 6 Proportion of gunny bags with breaking strength NOT between 3 and 7.5 kg per sq cm

Conclusion:

- We are using Cumulative Distribution Function (CDF) for finding the breaking strength NOT between 3 and 7.5 kg per sq cm
- From the above plot, we need to find out the probability or area under the purple-coloured lines for the given question that the proportion is NOT between 3 and 7.5 kg per sq. cm.
- We can see that the cumulative probability is 13.9%

Problem 3:

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level);

Data Observation:

- Checked first five and last five rows of the data Zingaro stone printing company.
- There are 75 rows and 2 columns in the Zingaro stone printing company dataset.
- The data types of both the columns are float64, no null values and the dataset has used 1.3 KB of storage.
- There is neither null values nor duplicates in the dataset Zingaro stone printing company.

Information about the dataset:

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 75 entries, 0 to 74
Data columns (total 2 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unpolished             75 non-null     float64
1   Treated and Polished   75 non-null     float64
dtypes: float64(2)
memory usage: 1.3 KB

```

Figure 7 data information

Statistical summary of the dataset:

	count	mean	std	min	25%	50%	75%	max	
Unpolished	75.000	134.111	33.042	48.407	115.330	135.597	158.215	200.161	
Treated and Polished	75.000	147.788	15.587	107.524	138.268	145.721	157.373	192.273	

Figure 8 Statical summary

Observations:

Mean:

- The mean value for treated and polished items (147.79) is higher than that of unpolished items (134.11).
- This indicates that, on average, treated and polished items are valued higher than unpolished ones.

Dispersion:

- The standard deviation for unpolished items (33.04) is significantly higher than that for treated and polished items (15.59).
- This suggests that the values of unpolished items have more variability compared to treated and polished items.

Range:

- The range (difference between max and min) for unpolished items is 151.75, while for treated and polished items, it is 84.75.
- This further indicates that unpolished items have a wider range of values.

Quartiles:

- The 25th and 75th percentiles for treated and polished items are 138.27 and 157.37, respectively, which shows a narrower interquartile range (IQR) compared to unpolished items (115.33 to 158.22).
- A narrower IQR for treated and polished items suggests more consistency in their values.

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

STEP 1:

HYPOTHESIS:

- **Null Hypothesis (H_0) :**
The mean Brinell's hardness index of unpolished stones is at least 150.
 $\Rightarrow H_0 : \mu \geq 150$
- **Alternate Hypothesis (H_1):**
The mean Brinell's hardness index of unpolished stones is less than 150.
 $\Rightarrow H_1 : \mu < 150$

STEP 2: Assumptions & Given:

- It is given that there is 5% significance level. So $\alpha = 0.05$
- Sample average = 134.1105
- Standard deviation of sample: 32.82
- Degree of freedom: 74
- It's not specified whether the population standard deviation is known, so we assume it is not.
- The dataset is having 75 rows which is $75 \geq 30$.
- $\mu = 150$
- The sample size = 75
- Critical value is: 1.96

STEP 4: Performing: t – test statistic

- We will conduct a one-sample t-test to determine if the mean hardness of the unpolished stones is significantly less than 150.
- There will be left- one – tailed – t-test, as we are proving the hypothesis test for the value which is less than the Bernell's hardness index.

$$\Rightarrow t - test = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

P value = 4.17128

One-Sample t-Test for Unpolished Stones

- The p-value is less than 0.05, We reject the null hypothesis and conclude that the unpolished stones may not be suitable for printing.
- We can say that unpolished stones do not have Brinell's hardness index even at least 150.

One-Sample t-Test for Unpolished Stones:
The t-statistic value: -4.164629601426757
The p-value: 4.171286997419652e-05
Reject the null hypothesis: The unpolished stones may not be suitable for printing.

Figure 9 One sample t test for unpolished stones

Conclusions:

- Compare the p-value with the significance level (0.05). If p-value < 0.05, reject the null hypothesis.
- We conclude that Zingaro may be justified in thinking that the current batch of unpolished stones may not be suitable for printing, as there is sufficient evidence to suggest that the mean hardness of unpolished stones is less than 150. Hence, we reject the Null hypothesis.

3.2 Is the mean hardness of the polished and unpolished stones the same?

STEP 1:

HYPOTHESIS:

- **Null Hypothesis (H_0) :**
The mean Brinell's hardness index of polished and unpolished stones is the same.
 $\Rightarrow H_0 : \mu_{polished} = \mu_{unpolished}$
- **Alternate Hypothesis (H_1):**
The mean Brinell's hardness index of polished and unpolished stones is different.
 $\Rightarrow H_1 : \mu_{polished} \neq \mu_{unpolished}$

STEP 2: Assumptions & given:

- It is given that there is 5% significance level. So $\alpha = 0.05$
- It's not specified whether the population standard deviation is known, so we assume it is not.
- The dataset is having 75 rows which is $75 \geq 30$.
- $\mu = 150$
- sample size = 75

STEP 3: Perform: Two – Tailed Sample T – Statistic -Test

- We have two samples and their variance are not equal.
- Both the samples are having same size
- Samples are continuous and independent in nature
- Here, $n > 30$. So here we will use 2 sample t-test.
- Variables are normally distributed.

Two-Sample t-Test for Polished and Unpolished Stones:
The t-statistic value: -3.2422320501414053
The p-value: 0.0014655150194628353
Reject the null hypothesis: The mean hardness of polished and unpolished stones is different.

Figure 10 2-sample t -test for mean hardness

STEP 4: Conclusion:

Here, we have conducted the 2-sample T – test for comparing the means of both the stones I.e. polished and unpolished stones: the p- value are: 0.0014 and t -stat value: -3.24

- p – value is less than 0.05, ($p - value < 0.05$) , So we are 95% confident to reject the null hypothesis H_0 . We have concluded that the mean hardness pf polished and unpolished stones are different.

Problem 4:

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may Favor one method above another and may work better in his/her favourite method. The response is the variable of interest.

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Dentist      90 non-null    int64
1   Method       90 non-null    int64
2   Alloy        90 non-null    int64
3   Temp         90 non-null    int64
4   Response     90 non-null    int64
dtypes: int64(5)
memory usage: 3.6 KB
```

Figure 11 Information about the dental hardness dataset

- The Data has 90 rows, indexed from 0 to 89 and there are 5 columns in the dataset.
- **Dentist:** Non-Null Count: 90, Data type: int64. This column contains int values which represents different dentists, and there are no missing values.
- **Method:** Non-Null Count: 90, data type: int64, this column contains int values representing different methods used, and all entries have no missing value.
- **Alloy:** Non-Null Count: 90, Data type: int64, this column contains integer values representing different alloys used, and all entries are non-null.
- **Temp:** Non-Null Count: 90, Data type: int64, this column contains integer values representing the temperatures at which processes were conducted, and all entries are non-null.
- **Response:** Non-Null Count: 90, Data type: int64, this column contains integer values representing the Brinell's hardness index, and all entries are non-null.
- **Memory Usage:** 3.6 KB.

Unique Value checks:

```
The Unique value of Dentist are : [1 2 3 4 5]
.....
The Unique value of Method are : [1 2 3]
.....
The Unique value of Alloy are : [1 2]
.....
The Unique value of Temp are : [1500 1600 1700]
.....
```

Figure 12 Checking unique value

Statistical Summary for the dataset:

	count	mean	std	min	25%	50%	75%	max
Dentist	90.000	3.000	1.422	1.000	2.000	3.000	4.000	5.000
Method	90.000	2.000	0.821	1.000	1.000	2.000	3.000	3.000
Alloy	90.000	1.500	0.503	1.000	1.000	1.500	2.000	2.000
Temp	90.000	1600.000	82.107	1500.000	1500.000	1600.000	1700.000	1700.000
Response	90.000	741.778	145.768	289.000	698.000	767.000	824.000	1115.000

Figure 13 Statistical Summary for the dataset:

Observations:

- **Dentist:**
Dentist column ranges from 1 to 5 and their mean is 3. From this we can say that the dentist is roughly even distribution across these values.
- **Method:**
The method variable are ranges from 1 to 3, with mean of 2. It suggests that a balanced use of the methods.
- **Alloy:**
The Alloy column has values of Alloy 1 and alloy 2 and the mean is 1.5. It indicates that there is an equal distribution between these two alloys.
- **Temp:**
The temperature are ranges from 1500 to 1700 with the mean of 1600. It suggests that the temperature which are being used are very close to the mean of temperature.
- **Response:**
The response variable is representing the Brinell's hardness index. The range of response are from 289 to 1115 with the mean of 741.778. This indicates the variability in hardness across the data.

visual analysis of the Dentist for the Alloy:

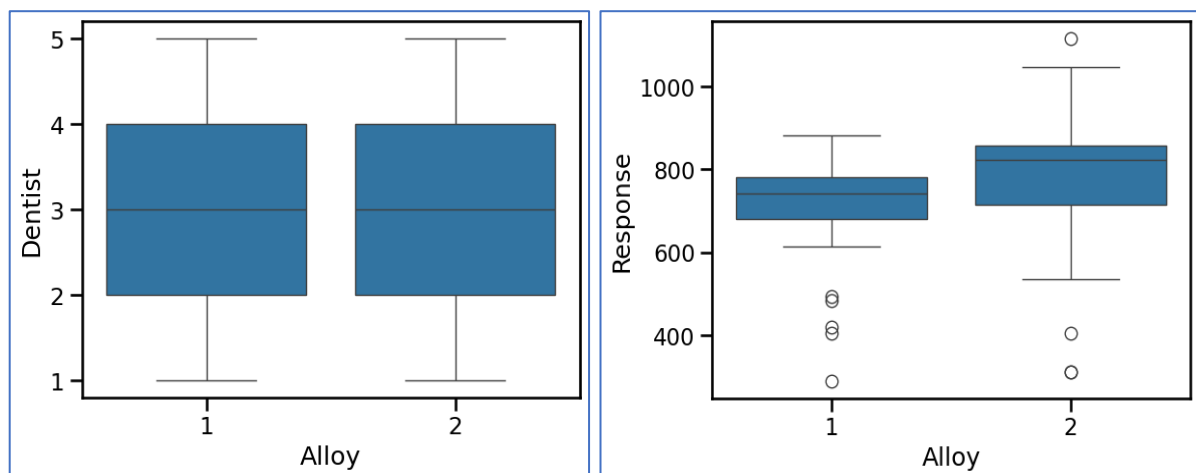


Figure 14 Visual analysis for dentist vs alloy & Response Vs Alloy

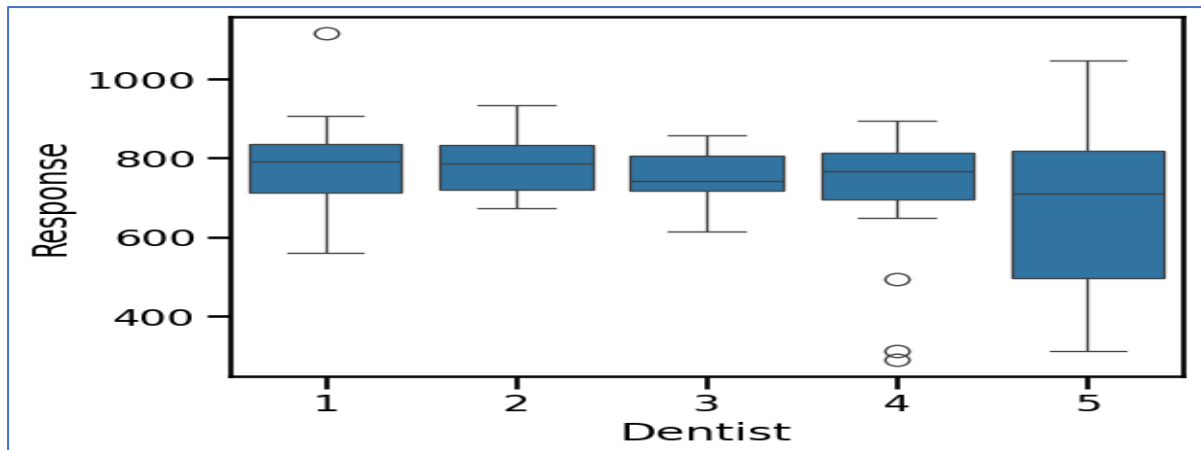


Figure 15 Dentist vs response

Observation:

- Dentist 1 and dentist 5 plot shows the distribution of response values.
- Dentist 5 has the widest spread which indicates the highest variability in response. It also shows no of outliers.
- Dentist 3 has smaller spread which indicates less variability in response values.
- The Median are relatively similar across all dentists, it indicates there is a difference.
- Dentist 2 and dentist 3 have slightly higher medians as compare to others.
- There are some outliers in the dataset especially for dentist 1, dentist 4 and dentist 5. It suggests that there are some extreme values present in the response distribution.

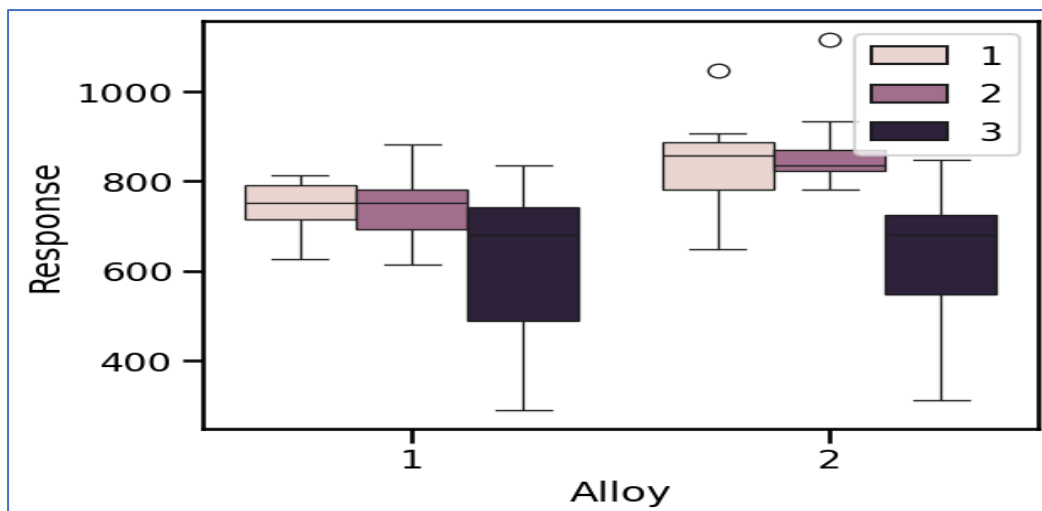


Figure 16 Multivariate graph

Observation:

For Alloy 1: The median response values for methods 1 and 2 are similar, while method 3 has a lower response. The IQR for method 1 is narrow, indicating less variability, while method 3 has a wider IQR, indicating more variability.

For Alloy 2: The median response values for methods 1 and 2 are slightly different, with method 3 again having a lower median response. The IQRs are similar in size across the methods, but the overall range and presence of outliers indicate some variability.

Conclusion:

This box plot helps in comparing the Brinell's hardness index across both the alloys and their methods. For Alloy 1 and Alloy 2, method 3 tends to have lower median as compare to method 1 and method 2. There are also few outliers present in the response column, especially in Alloy 2. Also there is some variability.

4.1 How does the hardness of implants vary depending on dentists?

Step 1: State the Hypothesis for both the Alloy:

FOR ALLOY 1:

- **Null Hypothesis (H_0) (For Alloy 1):**
There is no significant difference in the hardness of implants among the dentists for alloy 1.
- **Alternate Hypothesis (H_1) (For Alloy 1):**
There is a significant difference in the hardness of implants among the dentists for alloy 1.

STEP 2: Check the assumptions of the hypothesis test:

- **Normality:** We will check the residuals should be approximately distributed.
- **Homogeneity of variances:** The homogeneity of variance with in each group should be more or less equal.
- **Independence:** All the observations should be independent to each other

Step 3: Conducting Hypothesis test so that we can find the p value:

```
ANOVA Results for Alloy 1:
      sum_sq   df      F  PR(>F)
C(Dentist) 106683.689  4.000  1.977  0.117
Residual   539593.556 40.000   NaN    NaN
Shapiro-Wilk Test for Normality (Alloy 1) are : 0.0017434853361919522
Levene's Test for Homogeneity of Variances (Alloy 1): 0.2565537418543795

Fail to reject the null hypothesis: There is no significant difference in case of hardness of implants among the dentists for Alloy 1.
```

Figure 17 ANNOVA result for Alloy 1

After conducting the Shapiro Wilk Test and Homogeneity of variance, The ANOVA results are:

Test Results:

1. Sum of Squares (Within Groups): 539,593.56
2. Degrees of Freedom (Between Groups): 4
3. Degrees of Freedom (Within Groups): 40
4. F- Statistic test = 1.977
5. p-value: 0.117

Conclusion: FOR ALLOY 1:

The p- value is 0.117 which is greater than the significance level (0.05), So with this p value, we are very confident to that we fail to reject the Null Hypothesis. It shows that there is no significant difference in case of hardness of implants among the dentists for Alloy 1.

FOR ALLOY 2:

- **Null Hypothesis (H_0) (For Alloy 2):**

There is no significant difference in the hardness of implants among the dentists for alloy 2.

- **Alternate Hypothesis (H_1) (For Alloy 2):**

There is a significant difference in the hardness of implants among the dentists for alloy 2.

Step 2: Check the Assumptions of the Hypothesis Test:

We will conduct an ANOVA test for each alloy type. The assumptions for ANOVA are:

- The samples are independent.
- The populations are normally distributed.
- The populations have equal variances (homogeneity of variances).

STEP 3: Perform the Hypothesis test so that we can find the p value:

```
ANOVA Results for Alloy 2:
      sum_sq    df      F  PR(>F)
C(Dentist)  56797.911  4.000  0.525   0.718
Residual  1082204.889 40.000   NaN    NaN
Shapiro-Wilk Test for Normality (Alloy 2): 0.0076433937065303326
Levene's Test for Homogeneity of Variances (Alloy 2): 0.23686777576324952

Fail to reject the null hypothesis: There is no significant difference in case of hardness of implants among the dentists for Alloy 2.
```

Figure 18 ANNOVA Results for Alloy 2

Test Results:

1. Sum of Squares (Between Groups): 56,797.91
2. Sum of Squares (Within Groups): 1,082,205.00
3. Degrees of Freedom (Between Groups): 4
4. Degrees of Freedom (Within Groups): 40
5. F-Statistic: 0.525
6. p-value: 0.718

Conclusion:

From the test result, we found that the p – value is 0.718 which is greater than the level of significance value i.e. 0.05, So here also, we fail to reject the Null Hypothesis. Here also we have noticed that there is no significant difference in the hardness of implants among the dentists for Alloy 2.

4.2 How does the hardness of implants vary depending on methods?

Here, we will analyse the data, how hardness of dental implants varies depending upon the methods that is used for each type of alloy.

STEP 1: State the null and alternate hypotheses:

- **Null Hypothesis (H_0) (For Method):**

There is no significant difference in the mean hardness of dental between different methods.

$$\Rightarrow H_0 : \mu_{M_1} = \mu_{M_2} = \mu_{M_3} = \dots = \mu_k$$

- **Alternate Hypothesis (H_1) (For Method):**

There is significant difference in the mean hardness of dental between different methods.

$$\Rightarrow H_1 : \mu_{M_i} \neq \mu_{M_j} \text{ for some } i \neq j$$

STEP 2: Assumptions:

We will check for assumptions for conducting ANOVA

- **Normality:** We will find with the help of Shapiro – Wilk test.
- **Homogeneity of Variances:** We will use Levene’s test.
- **Independence:** Based on the study design, we assume that the observations are independent in nature.

STEP 3: Conduct the Hypothesis Test and Compute the p-value

Here, we will perform ONE-WAY -ANOVA for Each alloy for testing about the significance difference in mean hardness among the methods. If the results of ANOVA will be significant, then we go into deep to conduct the Tukey’s HSD test for pair-wise comparisons.

Analysis for Alloy 1:

- Filtering the data for both the alloy
- Checks for normality and homogeneity of variances
- Conduct the ANOVA Test.

```
Alloy 1 - Normality test p-value: Method
1    0.182
2    0.903
3    0.143
Name: Response, dtype: float64
Alloy 1 - Levene's test p-value: LeveneResult(statistic=6.52140454403598, pvalue=0.0034160381460233975)
Alloy 1 - ANOVA p-value:

```

	sum_sq	df	F	PR(>F)
C(Method)	148472.178	2.000	6.263	0.004
Residual	497805.067	42.000	NaN	NaN

```
Reject the null hypothesis: There is significant difference in the mean hardness of dental between different methods.
```

Figure 19 Alloy 1 checks for normality and homogeneity

Analysis for Alloy 1 (Result):

A. Normality Test:

Method 1: p-value = 0.182

Method 2: p-value = 0.903

Method 3: p-value = 0.143

Observation:

- Here, we have observed that all the method's p-value are greater than the significance value i.e. 0.05. So, from this we fail to reject the Null Hypothesis of the Shapiro-Wilk test.
- The dataset for each methods looks to be normally distributed

B. Homogeneity of Variances for Alloy 1:

Levene's test: p-value = 0.003

Observations:

- The p-value in case of levene's test are less than the level of significance i.e. 0.05. So, it indicates that the assumption of Homogeneity of Variances is failed.

C. ANOVA TEST:

F-statistic test: 6.263

p-value: 0.004

Observations:

- After analysis of ANOVA, we get the p-value which is less than the level of significance i.e. ($0.004 < 0.05$). so, from this we reject the Null Hypothesis. It Suggests that there is significant difference in mean hardness of implants between different method for alloy 1.

D. STEP 4: Tukey's HSD Test:

Since the ANOVA test is significant, we performed Tukey's HSD test to determine which pairs of methods differ.

```
<statsmodels.sandbox.stats.multicomp.TukeyHSDResults at 0x7bc8b05aebf0>
```

Figure 20 Tukey's HSD Test

```
The Tukey HSD Test Summary are : Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
1 2 -6.1333 0.987 -102.714 90.4473 False
1 3 -124.8 0.0085 -221.3807 -28.2193 True
2 3 -118.6667 0.0128 -215.2473 -22.086 True
-----
```

Figure 21 Tukey's HSD test

STEP 5: Conclusion for Alloy 1:

1. There is no significant difference in the mean hardness between both the methods i.e method 1 and method 2 (p-adj = 0.987)
2. There is no significant difference in mean hardness between Method 1 and Method 3. ((p-adj = 0.0085)
3. There is significant difference in mean hardness between Method 2 and Method 3. (p-adj = 0.0128)

Conclusion from the test results:

Normality: The normality assumption is reasonably met for each, method

Homogeneity of variances: Here in this case the assumption is violated.

ANOVA: Here there is significant difference in the mean hardness among the methods (p-value = 0.004.

Tukey's HSD Test Result:

Method 1 vs. Method 2: There is no significant difference

Method 1 vs. Method 3: There is significant difference

Method 2 vs. Method 3: There is significant difference

Based on the results above, method 3 has a significantly different impact on the hardness level for Alloy 1.

FOR ALLOY 2:

STEP 1: State the null and alternate hypotheses:

- **Null Hypothesis (H_0) (For Method):**

There is no significant difference in the mean hardness of dental between different methods.

$$\Rightarrow H_0 : \mu_{M_1} = \mu_{M_2} = \mu_{M_3} = \dots = \mu_k$$

- **Alternate Hypothesis (H_1) (For Method):**

There is significant difference in the mean hardness of dental between different methods.

$$\Rightarrow H_1 : \mu_{M_i} \neq \mu_{M_j} \text{ for some } i \neq j$$

STEP 2: Assumptions:

We will check for assumptions for conducting ANOVA

- **Normality:** We will find with the help of Shapiro – Wilk test.
- **Homogeneity of Variances:** We will use Levene's test.
- **Independence:** Based on the study design, we assume that the observations are independent in nature.

STEP 3: Conduct the Hypothesis Test and Compute the p-value

Here, we will perform ONE-WAY -ANOVA for Each alloy for testing about the significance difference in mean hardness among the methods. If the results of ANOVA will be significant, then we go into deep to conduct the Tukey's HSD test for pair-wise comparisons.

Analysis for Alloy 2:

- Filtering the data for both the alloy
- Checks for normality and homogeneity of variances
- Conduct the ANOVA Test.

```
Alloy 2 - Normality test p-value: Method
1  0.758
2  0.001
3  0.103
Name: Response, dtype: float64
Alloy 2 - Levene's test p-value: LeveneResult(statistic=3.349707184158617, pvalue=0.04469269939158668)
Alloy 2 - ANOVA p-value:
C(Method) 499640.400  2.000 16.411  0.000
Residual  639362.400 42.000   NaN    NaN

Reject the null hypothesis: There is significant difference in the mean hardness of dental between different methods.
```

Figure 22 Alloy 2 checks for normality and homogeneity

Analysis for Alloy 2 (Result):

A. Normality Test:

Method 1: p-value = 0.758

Method 2: p-value = 0.001

Method 3: p-value = 0.103

Observation:

- Here, we have observed that p- value of method 1 and method 3 are greater than the significance value i.e. 0.05. So, from this we fail to reject the Null Hypothesis of the Shapiro-Wilk test.
- The dataset for each methods looks to be normally distributed.

B. Homogeneity of Variances:

Levene's test: p-value = 0.044

Observations:

- The p-value in case of levene's test are less than the level of significance i.e. 0.05. So, it indicates that the assumption of Homogeneity of Variances is failed.

C. ANOVA TEST:

F-statistic test: 16.411

p-value: 0.00014

Observations:

- After analysis of ANOVA, we get the p- value which is less than the level of significance i.e. (0.00014 < 0.05). so, from this we reject the Null Hypothesis. It Suggests that there is significant difference in mean hardness of implants between different method for alloy 2.

D. Tukey's HSD Test:

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1	2	27.0	0.8212	-82.4546	136.4546	False
1	3	-208.8	0.0001	-318.2546	-99.3454	True
2	3	-235.8	0.0	-345.2546	-126.3454	True

Figure 23 Tukey's HSD Test

STEP 5: Conclusion for Alloy 2:

1. There is a significant difference in the mean hardness between both the methods i.e method 1 and method 2 (p-adj = 0.08212)
2. There is no significant difference in mean hardness between Method 1 and Method 3. (p-adj = 0.0001).
3. There is no significant difference in mean hardness between Method 2 and Method 3. (p-adj = 0.00)

Conclusion from the test results:

Normality: The normality assumption is reasonably met for each, method

Homogeneity of variances: Here in this case the assumption is violated.

ANOVA: Here there is significant difference in the mean hardness among the methods (p-value = 0.00014)

Tukey's HSD Test Result:

Method 1 vs. Method 2: There is no significant difference

Method 1 vs. Method 3: There is significant difference

Method 2 vs. Method 3: There is significant difference

Based on the results above, method 3 has a significantly different impact on the hardness level for Alloy 2.

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

For Alloy 1:

Null Hypothesis (H_0) (For Alloy 1):

There is no significant difference among the interaction effect between dentist and method on the hardness of dental implants for alloy 1.

Alternate Hypothesis (H_1) (For Alloy 1):

There is significant difference among the interaction effect between dentist and method on the hardness of dental implants for alloy 1.

- I. The x-axis can represent the method.
- II. The y-axis can represent the hardness (response).
- III. Use different lines for different dentists.

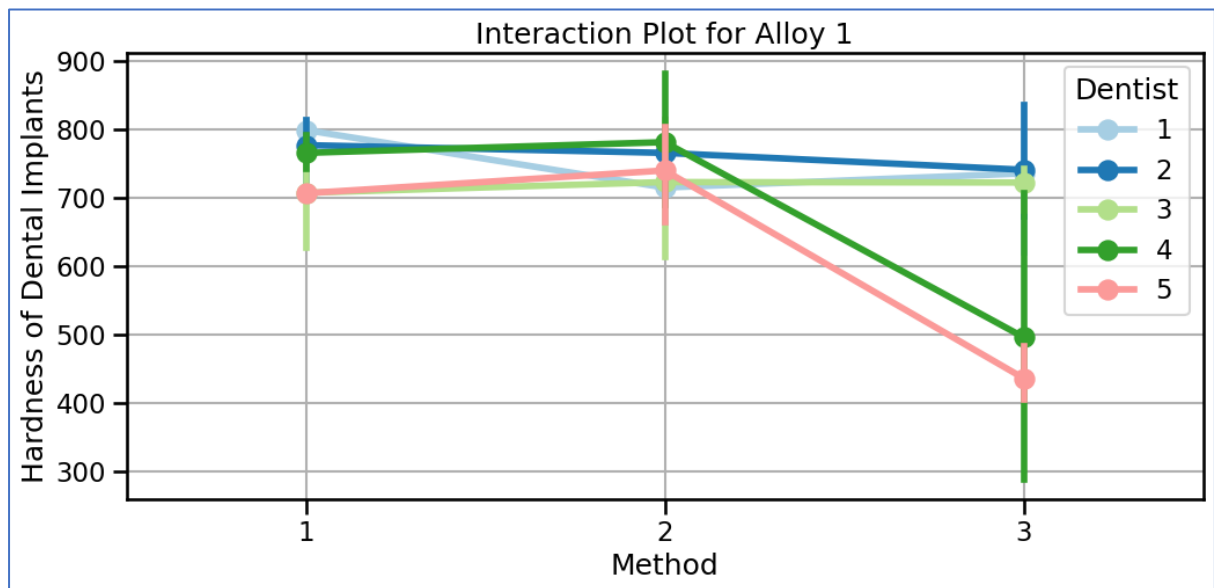


Figure 24 Interaction plot for Alloy 1

Inferences from the Interaction Plots:

Alloy 1:

Dentist Influence:

All the dentists showing the different hardness response across all the methods.

Dentist number 4 showing the result differs more significantly across all the methods as compare to other dentist.

Method Influence:

Method number 1 shows the higher hardness response value than the method 2 for almost dentist.

Interaction Effect:

When we look for lines which is not parallel, that indicates an interaction effect between dentist and method.

Example: Dentist 2 has relatively consistent hardness response across the methods, where in case of Dentist 1, shows more significant drop in hardness response from method 1 to method 2.

Conclusion:

The p-value is 0.000003 which is very less as compare to alpha i.e. the level of significance (0.05). Hence, we reject the Null hypothesis and suggests that There is significant difference among the interaction effect between dentist and method on the hardness of dental implants for alloy 1.

For Alloy 2:

Null Hypothesis (H_0) (For Alloy 2):

There is no significant difference among the interaction effect between dentist and method on the hardness of dental implants for alloy 2.

Alternate Hypothesis (H_1) (For Alloy 2):

There is significant difference among the interaction effect between dentist and method on the hardness of dental implants for alloy 2.

- I. The x-axis can represent the method.
- II. The y-axis can represent the hardness (response).
- III. Use different lines for different dentists.

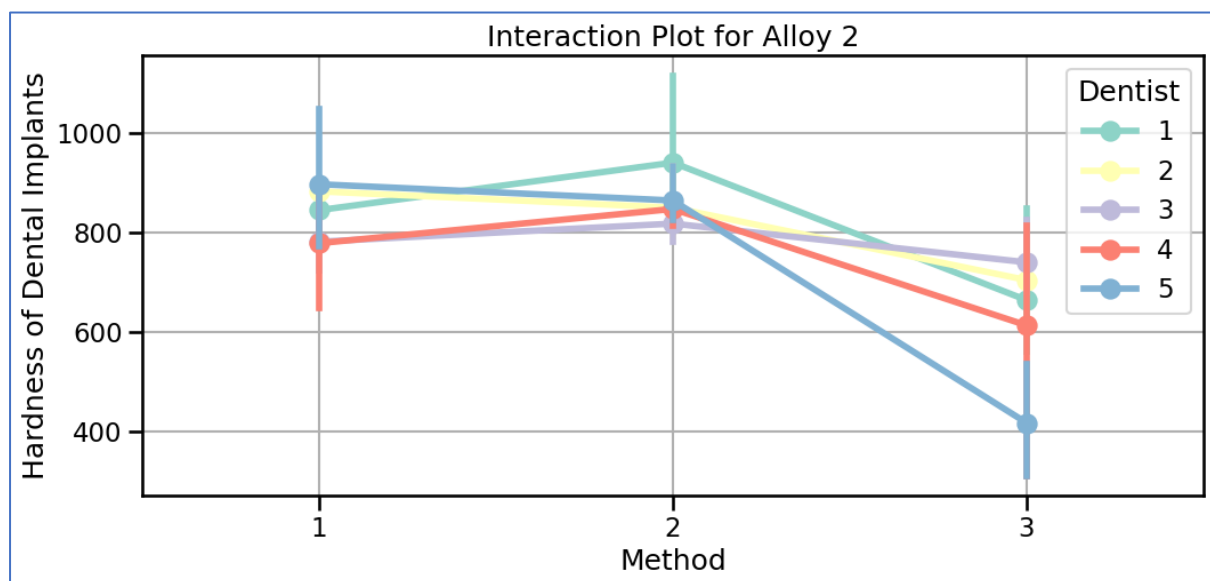


Figure 25 Interaction plot for Alloy 2

Inferences from the Interaction Plots:

Alloy 2:

Dentist Influence:

All the dentists showing varying in hardness response across all the methods. Dentist number 4 showing the result which is having distinct pattern as compare to other dentist.

Method Influence:

Method 1 and method 2 in alloy 2 shows more variability in hardness of response across different dentists. In case of method 2 which is showing the lower hardness of response values.

Interaction Effect:

The interaction effect is more pronounced in Alloy 2. Number 1 and Number 2 dentists shows more consistent result across all the methods, while Dentist number 3 and dentist number 4 exhibit greater variability.

Conclusion:

The p-value is 0.000074 which is very less as compare to alpha i.e. the level of significance (0.05). Hence, we reject the Null hypothesis and suggests that There is significant difference among the interaction effect between dentist and method on the hardness of dental implants for alloy 1.

Summary: The interaction effect between the dentist and method on the hardness of dental implants for each type of alloy

- The interaction plot suggests towards a significant relation between the dentist and the methodology used, which means that how effective a method is varies depending on the dentist.
- Both of the alloys represented different patterns of hardness, while alloys 1 is showing more steady results across methods and dentist when compared to alloy 2

4.4 How does the hardness of implants vary depending on dentists and methods together?

STEP 1: State Hypothesis:**TWO WAY ANOVA FOR ALLOY 1:**

- **Null Hypothesis (H_0) (For Alloy 1):**
 1. The Mean hardness of dental implant is same for all the methods.
 2. The mean hardness of dental implants is the same for all dentist.
 3. The is no interaction effect between the method and the dentist on the hardness of dental implants.
- **Alternate Hypothesis (H_1) (For Alloy 1):**
 1. The Mean hardness of dental implant is different for all the methods.
 2. The mean hardness of dental implants is the different for all dentist.
 3. The is an interaction effect between the method and the dentist on the hardness of dental implants.

Step 2: Check for Assumptions:

1. **Normality:** Here we will use Shapiro-wilk test, for checking the residual should be normally distributed.
2. **Homogeneity of variances:** The variances for each group should be approximately equal
3. **Independence:** The observation should be independent of each other

STEP 3: Perform Hypothesis Test to find the p-value :

```

The results of Alloy 1 are :
(
      sum_sq      df      F      PR(>F)
C(Dentist)      106683.689    4.000    3.900    0.011
C(Method)      148472.178    2.000   10.854    0.000
C(Dentist):C(Method) 185941.378    8.000    3.398    0.007
Residual      205180.000   30.000      NaN      NaN,
ShapiroResult(statistic=0.9565479159355164, pvalue=0.09016714990139008),
LeveneResult(statistic=1.2383447671231802, pvalue=0.26882091488547033))

```

Figure 26 Alloy 1 ANOVA Analysis

ANOVA Table for Alloy 1:

Source	Sum of Squares	df	F-value	p-value
Dentist	106,683.69	4	3.90	0.011
Method	148,472.18	2	10.85	0.0003
Dentist	185,941.38	8	3.40	0.0068
Residual	205,180.00	30		

Observation: Assumptions for Alloy 1:

- **Interaction Effect:** The dentist interaction effect is: 0.0068
- **Normality Test:** Shapiro-Wilk p-value = 0.0902 (not significant, normality assumption holds)
- **Homogeneity of Variances Test:** Levene's Test p-value = 0.2688 (not significant, homogeneity assumption holds)

Tukey HSD test:

To identify which dentists and methods combinations are different, we will perform post-hoc tests for Alloy 1.

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1:1	1:2	-84.0	0.9933	-332.8283	164.8283	False
1:1	1:3	-63.3333	0.9996	-312.1617	185.495	False
1:1	2:1	-22.0	1.0	-270.8283	226.8283	False
1:1	2:2	-33.3333	1.0	-282.1617	215.495	False
1:1	2:3	-58.0	0.9999	-306.8283	190.8283	False
1:1	3:1	-91.6667	0.9853	-340.495	157.1617	False
1:1	3:2	-76.0	0.9975	-324.8283	172.8283	False
1:1	3:3	-76.6667	0.9972	-325.495	172.1617	False
1:1	4:1	-33.3333	1.0	-282.1617	215.495	False
1:1	4:2	-17.6667	1.0	-266.495	231.1617	False
1:1	4:3	-302.6667	0.007	-551.495	-53.8383	True
1:1	5:1	-92.3333	0.9844	-341.1617	156.495	False
1:1	5:2	-59.0	0.9998	-307.8283	189.8283	False
1:1	5:3	-362.6667	0.0007	-611.495	-113.8383	True
1:2	1:3	20.6667	1.0	-228.1617	269.495	False
1:2	2:1	62.0	0.9997	-186.8283	310.8283	False
1:2	2:2	50.6667	1.0	-198.1617	299.495	False
1:2	2:3	26.0	1.0	-222.8283	274.8283	False
1:2	3:1	-7.6667	1.0	-256.495	241.1617	False
1:2	3:2	8.0	1.0	-240.8283	256.8283	False

Figure 27 Tukey Test

Specific Results

- **False:** There is not enough evidence to reject the null hypothesis at the 5% significance level. The adjusted p-value (p-adj) is greater than 0.05.
- **True:** There is enough evidence to reject the null hypothesis at the 5% significance level. The adjusted p-value (p-adj) is less than or equal to 0.05.

Observation:

- The pairs which are assigned to True in the reject column have a significant difference in their means at 5% significant level.
- The pairs which are assigned to False in the reject column do not have a significant difference in their means at 5% significant level.

Conclusion: For Alloy 1:

- **Interaction Effect:** There is a significant interaction effect between the dentists and the methods on the hardness of the implants (p-value = 0.0068).
- **Main Effect:** The dentists and the methods both are having significant main effects on the hardness of the implants (p-value <0.05)

FOR ALLOY 2:

STEP 1: State Hypothesis:

- **Null Hypothesis (H_0) (For Alloy 2):**
 1. There is no interaction effect between dentists and methods on the hardness of implants.
- **Alternate Hypothesis (H_1) (For Alloy 2):**
 1. There is an interaction effect between dentists and methods on the hardness of implants.

Step 2: Check for Assumptions:

4. **Normality:** Here we will use Shapiro-wilk test, for checking the residual should be normally distributed.
5. **Homogeneity of variances:** The variances for each group should be approximately equal
6. **Independence:** The observation should be independent of each other

STEP 3: Perform Hypothesis Test to find the p-value :

```
The results of Alloy 2 are :
(
      sum_sq      df      F  PR(>F)
C(Dentist)      56797.911  4.000  1.106  0.372
C(Method)      499640.400  2.000 19.461  0.000
C(Dentist):C(Method) 197459.822  8.000  1.923  0.093
Residual      385104.667 30.000   NaN   NaN,
ShapiroResult(statistic=0.9570877552032471, pvalue=0.09455247968435287),
LeveneResult(statistic=0.28029313901247516, pvalue=0.5978422265057014))
```

Figure 28 ANOVA on Alloy 2

ANOVA table for Alloy 2:

Source	Sum of Squares	df	F-value	p-value
Dentist	56,797.91	4	1.11	0.372
Method	499,640.40	2	19.46	0.000004
Dentist	197,459.82	8	1.92	0.093
Residual	385,104.67	30		

Observation: Assumptions for Alloy 2:

- **Interaction Effect (Dentist):** The interaction effect for the dentist is 0.093 (Not Significant)
- **Normality Test:** Shapiro-Wilk p-value = 0.0946, which is not significant (normality assumption holds)
- **Homogeneity of variances test:** The Levene's test p-value = 0.5978 which is not significant. So the homogeneity assumption holds.

Tukey's HSD Test:

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1:1	1:2	95.3333	0.999	-245.5625	436.2292	False
1:1	1:3	-180.6667	0.8085	-521.5625	160.2292	False
1:1	2:1	37.6667	1.0	-303.2292	378.5625	False
1:1	2:2	5.0	1.0	-335.8958	345.8958	False
1:1	2:3	-140.3333	0.9635	-481.2292	200.5625	False
1:1	3:1	-63.3333	1.0	-404.2292	277.5625	False
1:1	3:2	-27.0	1.0	-367.8958	313.8958	False
1:1	3:3	-104.6667	0.9973	-445.5625	236.2292	False
1:1	4:1	-66.3333	1.0	-407.2292	274.5625	False
1:1	4:2	2.3333	1.0	-338.5625	343.2292	False
1:1	4:3	-231.3333	0.4686	-572.2292	109.5625	False
1:1	5:1	52.0	1.0	-288.8958	392.8958	False
1:1	5:2	19.3333	1.0	-321.5625	360.2292	False
1:1	5:3	-427.0	0.0049	-767.8958	-86.1042	True
1:2	1:3	-276.0	0.2169	-616.8958	64.8958	False
1:2	2:1	-57.6667	1.0	-398.5625	283.2292	False

Figure 29 Tukey's HSD test for Alloy 2

Specific Results:

- **Significant Difference:** The only significant difference is between dentists 1:1 and 1:5 with a mean difference of 276.0, a p-value of 0.0049, and the 95% confidence interval not including zero. This is indicated by reject = True.
- **Non-Significant Differences:** All other comparisons show non-significant differences with reject = False, meaning the hardness of implants does not differ significantly between these pairs of dentists.

Observation: From this test, we can say that for Alloy 1, the hardness of implants is significantly different between dentists 1:1 and 1:5. For all other pairs of dentists, there is no significant difference in the hardness of implants.

Step 4: Conclusion:

Interaction Effect: There is no significant interaction effect between the dentists and the methods on the hardness of the implants (p-value = 0.093).

Main Effect: The effect of method is significant (p-value = 0.000004). but the effect of dentists is not significant (p-value = 0.372) which is (p-value >0.05)